



Comsats University Islamabad, Lahore Campus

Theory Assignment <4> – SPRING 2025

Course Title	Software Testing	Course Code	CSE455	Credit Hours		3(2,1)	
Instructor/s	Ms. Yella Mehroze	Program Name	BSSE				
Semester	8 th	Batch	FA21	Section	A	Date	June 16, 2025
Time Allowed	1 week	Maximum Marks					
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Question Statement

[CLO-4]; Bloom Taxonomy Level: <Analyzing>

Analyze your final year project and design the following reports

1. Test Summary Report
2. Defect Report

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Timber Trails Test Summary Report

Test Objects

Test object ID (TID)	Test Object Name
T_1	Loading Game
T_2	Navigating Main Menu
T_3	Controlling Player
T_4	Progressive Zones
T_5	Collecting Items/Power-ups
T_6	Avoiding Hurdles
T_7	Accomplishing Objectives
T_8	Combat System
T_9	Inventory Management
T_10	Challenge Mechanics

What was Actually Tested

System tested was performed by both team members while first checking major/core features specified in the requirement document. Integration of all the features was also tested in the game. At the end, game was played with end-user perspective to check for the lack of any major feature. Compatibility with platform was verified.

Dates

TID	Tested on
T_1	03/06/2025
T_2	05/06/2025
T_3	03/06/2025
T_4	11/06/2025
T_5	11/06/2025
T_6	12/06/2025
T_7	03/06/2025
T_8	06/06/2025
T_9	06/06/2025
T_10	12/06/2025

When the Tests were Performed

TID	TCID	Tested at	Platform
T_1	T1C_1	02:50 PM	Windows 10, 11
	T1C_2	03:23 PM	
	T1C_3	04:00 PM	
T_2	T2C_1	01:10 PM	Windows 10, 11
	T2C_2	12:00 PM	
T_3	T3C_1	01:45 PM	Windows 10, 11
	T3C_2	02:05 PM	
	T3C_3	02:55 PM	
T_4	T4C_1	01:05 PM	Windows 10, 11
	T4C_2	02:30 PM	
T_5	T5C_1	12:50 PM	Windows 10, 11
	T5C_2	01:50 PM	
	T5C_3	02:35 PM	
T_6	T6C_1	01:55 PM	Windows 10, 11
	T6C_2	02:45 PM	
	T6C_3	03:20 PM	
	T6C_4	04:08 PM	
T_7	T7C_1	02:50 PM	Windows 10, 11
	T7C_2	03:55 PM	
T_8	T8C_1	12:30 PM	Windows 10, 11
	T8C_2	01:20 PM	
	T8C_3	02:38 PM	
T_9	T9C_1	02:32 PM	Windows 10, 11
	T9C_2	03:10 PM	
T_10	T10C_1	04:50 PM	Windows 10, 11
	T10C_2	05:30 PM	

Summary

Timber Trails consist of two zones where the game play is termed as survival, exploration, and fighting. So, each of the zones has unique difficulties, enemies, environment, and mission tasks depending on what level in the game we want to open. Pros: Only those items and several parts of the puzzles must be collected to proceed to the new zone and all of them must complete mini games which are also missions.

Testing is performed considering each level, feature. Looking at available time, most important features are test first and then the less important one.

Types of Tests and Focus

1. Unit Testing

Unit testing is the process of testing individual system units. Smallest components are tested individually. It can be performed using automated tools or using manual techniques as well.

- Loading Game Unit
- Main Menu Sub-Unit
- Mini Challenge Unit
- Power-up Collection Unit
- Gameplay Unit

2. Integration Testing

Modules are combined and then tested to verify the compatibility of each individual component with other components. Modules are the units already tested in unit testing. In this testing phase, we are going to test the logic or actions that player will be able to perform.

- Character Selection
- Audio Settings
- Avoiding Hurdles
- Combat System
- Inventory Management

3. System Testing

System testing is part of black box testing in which the entire system is tested in search of any error. This test is performed to ensure proper working of the entire system according to the requirements. It tests functionality from an end-to-end perspective.

4. Acceptance Testing

It is the verification of the system against documented requirements to ensure that no further alteration is required.

Test Progress Statistics

Planned	26 tests
Run	110
Blocked	5
Automated Test coverage	80%
Key blockers	1. API downtime 2. Missing UI components

Test Object Quality Statistics

New Defects	15
In Progress	6
Corrected	7
Comment(s)	Critical issues related to package compatibility with other zones similar packages are found and need immediate fix.

Risk

1. If the first round of testing is not completed within 1 week, it could delay bug fixes and final testing. If this happens, UAT would be pushed back and eventual affect the launch date.
2. If the members don't have a basic understanding of the feature they are testing, testing could be delayed or not conducted properly.
3. Testing should be paused immediately if either system experiences player movement impairment or any crash.

Deviation from Plan

Project was aimed to implement three distinct zones but due to the different factors like game satisfaction and time period. After implementing two zones, we have decided to skip 3rd zone but just improve existing two zones with enhanced challenge mechanics.

Forecast

Remaining 10 test cases to be executed in the next iteration. Continued regression testing and defect retesting planned.

Overall Assessment

High confidence in the stability of the character customization menu and help option. Further confirmation is needed for last zone challenges.

Timber Trails Defect Summary Report

1. Loading Game Defect Report

Classification	
Status	Closed
Class	UI/UX
Priority	Low
Root Cause	N/A – No defect identified
Discovery	During functional test on 04/06/2025
Problem description	
Test steps	1. Launch the game 2. Wait for the loading process to complete 3. Check for successful transition to main menu
Expected behavior	The system should display a loading screen with a visible progress indicator and then automatically transition to the main menu.
Actual behavior	All expected behaviors were observed correctly. No defect found in loading flow.
History	
Measures taken	Test case executed. No issue observed.
Comments	Consider enhancing the loading screen with animations.
Miscellaneous	Test Case ID: 01 Subsystem: User Design
Approval	Approved by: Muhammad Talha Shafique

2. Combat System Defect Report

Classification	
Status	Open
Class	I/O
Priority	High
Root Cause	Code assumed WASD input instead of navigation keys.
Discovery	During functional test on 05/06/2025
Problem description	
Test steps	1. The player engages in combat with an enemy using melee or ranged attacks. 2. The player uses defensive action such as dodging or blocking. 3. The enemy retaliates. 4. The player defeats the enemy. 5. The player's health reaches zero during combat.
Expected behavior	The enemy takes damage, and the health indicator for the enemy decreases.
Actual behavior	Health of enemy is changed inconsistently.
History	
Measures taken	Test case executed. Enemy health issues observed.
Comments	Consider enhancing the enemy controls.
Miscellaneous	Test Case ID: 02 Subsystem: Player Controls

Approval	Approved by: Aoun Haider
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3. Accomplishing Objectives Defect Report

Classification	
Status	Closed
Class	Environmental
Priority	Medium
Root Cause	N/A – No defect identified
Discovery	During functional test on 08/06/2025
Problem description	
Test steps	1. The player performs actions required for each objective (e.g., collecting items, defeating enemies). 2. The player completes all mandatory objectives. 3. The player fails to complete an objective within the zone's constraints (e.g., time limit).
Expected behavior	The system displays a message or animation indicating objective completion and progress to the next zone.
Actual behavior	A pop-up is displayed indicating completed challenge.
History	
Measures taken	Test case executed. No issue found.
Comments	N/A
Miscellaneous	Test Case ID: 03 Subsystem: Player Controls
Approval	Approved by: Aoun Haider

4. Avoiding Hurdles Defect Report

Classification	
Status	Closed
Class	Zone-Specific
Priority	Low
Root Cause	N/A – No defect identified
Discovery	During functional test on 11/06/2025
Problem description	
Test steps	1. The player approaches a hurdle. 2. The player attempts to jump or dodge to avoid the hurdle. 3. The player avoids the hurdle. 4. The player collides with the hurdle.
Expected behavior	The system continues gameplay without reducing health or speed and displays an animation confirming successful avoidance.
Actual behavior	Health is not changed.
History	
Measures taken	Test case executed. No issue found.
Comments	More engaging hurdles are required to add based on the zone.
Miscellaneous	Test Case ID: 04 Subsystem: Environmental
Approval	Approved by: Muhammad Talha Shafique

5. Collecting Items/Power-ups Defect Report

Classification	
Status	Closed
Class	UI/UX
Priority	Low
Root Cause	N/A – No defect identified
Discovery	During functional test on 05/06/2025
Problem description	
Test steps	1. Player approaches and collects an item/power-up in the mini-challenge. 2. Player checks inventory after collection. 3. The player uses a power-up.
Expected behavior	The system displays a countdown timer and applies the effect.
Actual behavior	Effect is clearly visible with a pop-up for entering the challenge.
History	
Measures taken	Test case executed. No issue found.
Comments	N/A
Miscellaneous	Test Case ID: 05 Subsystem: UI Controller
Approval	Approved by: Muhammad Talha Shafique

6. Progressive Zones Defect Report

Classification	
Status	Closed
Class	Environmental
Priority	Medium
Root Cause	N/A – No defect identified
Discovery	During functional test on 03/06/2025
Problem description	
Test steps	1. Players complete the current zone objectives. 2. Player proceeds to the new zone. 3. Player encounters zone-specific challenges and enemies.
Expected behavior	The system displays a transitional animation to indicate zone completion.
Actual behavior	A pop-up is displayed indicating zone completion.
History	
Measures taken	Test case executed. No issue found.
Comments	Animation or some visual effect is appropriate to enhance game interaction.
Miscellaneous	Test Case ID: 06 Subsystem: Environmental
Approval	Approved by: Aoun Haider

7. Controlling Player Defect Report

Classification	
Status	Closed
Class	Environmental/Exploratory
Priority	High

Root Cause	N/A – No defect identified
Discovery	During functional test on 06/06/2025
Problem description	
Test steps	1. The player presses directional keys to move the character. 2. The player presses the jump key. 3. A player interacts with an object (e.g., opens a door). 4. When player moves. 5. The player was stuck, so repositioned.
Expected behavior	The character moves left, right, forward, or backward accordingly.
Actual behavior	Character is moved consistently against corresponding key.
History	
Measures taken	Test case executed. No issue found.
Comments	N/A
Miscellaneous	Test Case ID: 07 Subsystem: Exploratory
Approval	Approved by: Aoun Haider

8. Navigating Main Menu Defect Report

Classification	
Status	Closed
Class	UI/UX
Priority	Medium
Root Cause	N/A – No defect identified
Discovery	During functional test on 01/06/2025
Problem description	
Test steps	1. Player navigates to the "Play Game" option using the input device. 2. Player selects the "Settings" option. 3. The player navigates and selects the "Help" option. 4. Player chooses the "Exit" option.
Expected behavior	The system highlights the "Play Game" option and starts the game.
Actual behavior	'Play Game' button is clearly visible and intractable.
History	
Measures taken	Test case executed. No issue found.
Comments	Help demo video needed to more intuitive.
Miscellaneous	Test Case ID: 08 Subsystem: UI/UX
Approval	Approved by: Aoun Haider

9. Challenge Mechanics Defect Report

Classification	
Status	Closed
Class	Exploratory
Priority	High
Root Cause	N/A – No defect identified

Discovery	During functional test on 12/06/2025
Problem description	
Test steps	1. The player encounters a challenge, such as a puzzle or trap. 2. The player engages with the challenge by interacting with it (e.g., solving a puzzle or avoiding a trap). 3. The player completes the challenge successfully. 4. The player fails to complete the challenge (e.g., time runs out or incorrect actions).
Expected behavior	The system displays a failure message, deducts health or resources as a penalty, and allows the player to retry if applicable.
Actual behavior	A pop-up message is displayed asking for retry to return back.
History	
Measures taken	Test case executed. No issue found.
Comments	Arrangement of challenges is not semantically appropriate and requires proper rearrangement.
Miscellaneous	Test Case ID: 09 Subsystem: Exploratory
Approval	Approved by: Muhammad Talha Shafique

10. Inventory Management Defect Report

Classification	
Status	Closed
Class	UI/UX
Priority	Low
Root Cause	N/A – No defect identified
Discovery	During functional test on 05/06/2025
Problem description	
Test steps	1. The player opens the Progress icon.
Expected behavior	The system displays a failure message, deducts health or resources as a penalty, and allows the player to retry if applicable.
Actual behavior	Entire zone progress is clearly visible in the triggered pop-up menu.
History	
Measures taken	Test case executed. No issue found.
Comments	N/A
Miscellaneous	Test Case ID: 10 Subsystem: UI/UX
Approval	Approved by: Aoun Haider